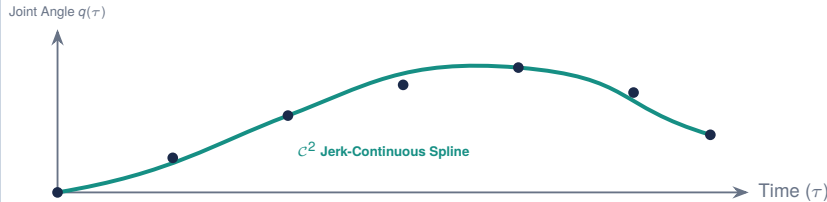




## 1. Overlapping Action Chunks at Successive Cycles ( $A^{(0)}, A^{(1)}, A^{(2)}$ )



## 2. Continuous Synthesized Trajectory $\mathbf{a}^*(\tau)$ Executed by MCU at 1000 Hz



### Exponential Temporal Ensembling

At physical time  $\tau$ , multiple overlapping chunks  $A^{(i)}$  provide candidate predictions  $\mathbf{a}_{\tau-t_i}^{(i)}$ . The blended command  $\mathbf{a}^*(\tau)$  is:

$$\mathbf{a}^*(\tau) = \frac{\sum_{i=0}^{N-1} w_i \cdot \mathbf{a}_{\tau-t_i}^{(i)}}{\sum_{i=0}^{N-1} w_i}$$

where weights decay exponentially with the age of the prediction:

$$w_i = \exp(-\lambda(\tau - t_i))$$

### Physical AI Systems Impact:

- **Noise Filtering:** Suppresses high-frequency diffusion sampling jitter.
- **Delay Amortization:** 50 ms inference latency is amortized across 320 ms of motion.
- **Continuity Guarantee:** Prevents discrete velocity jumps at chunk borders.

**Systems Performance Law:** Action chunking with temporal ensembling converts stochastic, discrete neural inference into a continuous reference reservoir. By executing  $K = 4$  steps before blending the next chunk, the real-time MCU never starves, maintaining 1000 Hz motor control with zero mechanical shock.